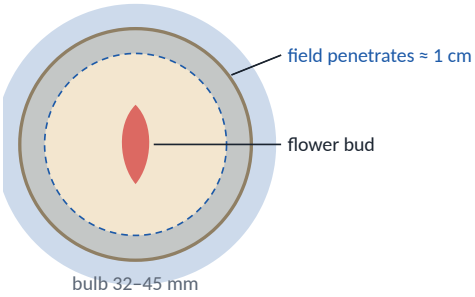
# Gentle microwave treatment of tulip bulbs

Parostok builds on research into microwave seed treatment since 2000. Together with Belgian partners we propose to develop a dedicated machine for chemical-free treatment of living tulip bulbs — designed first of all not to harm the bulb.

## Why a dedicated machine

- A tulip bulb (32–45 mm) is several times larger than the depth a 2.45 GHz field penetrates (about 1 cm).
- In round objects of this size microwave heating can concentrate at the centre — exactly where the flower bud sits.
- Seed equipment therefore cannot simply be reused: bulbs need their own handling, dosing and control.




## Design principles

- Single layer of rotating bulbs, no drops or impacts.
- Dose delivered in sections, with equalisation zones in between.
- **Hard limit: the bulb core never exceeds 42 °C.** Hot-water treatment above about 45 °C is considered harmful to tulip bulbs.
- Optional low-heat pulsed mode.
- Every batch logged: power, temperatures, recipe.
- Easy-to-clean contact surfaces, so that disease is not carried between lots.

### Where we stand

The effect of microwave treatment on tulip bulbs has not been demonstrated yet: we found no published study on living tulip bulbs. The programme therefore starts with bud-safety and pilot forcing trials against an untreated control, and stops if they show no benefit.

## What we are looking for in Belgium

- Growers and forcers who can provide bulbs and forcing space for trials.
- An independent research partner for the safety and forcing trials.
- End users and co-funding partners for the prototype.

## Staged programme

| | | | |
|---|---|---|---|
| 1 | Bulb properties, bud-safety threshold, pilot forcing → go / no-go | 6–9 months | **€21–74k** |
| 2 | Design and prototype machine (0.5 t/h) | about 2 years | **after gate 1** |
| 3 | Independent forcing and field trials | one season | **after phase 2** |

**Indicative total: up to about €0.5 million. Evidence-based result expected in 2029–2030.**

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